

Lecture 2

Inflation and the Phillips Curve

Bank of Portugal—Empirical Macroeconomics

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Overview

- Phillips Curve—basic facts and main questions
- Time series estimation
- Cross sectional / regional estimation
- **After lunch:** inflation after the Pandemic

Modern New Keynesian Phillips Curve

$$\pi_t = \beta E_t \pi_{t+1} + \kappa(y_t - y_t^n) + \eta_t$$

- Three drivers of inflation:
 - Expected inflation: $E_t \pi_{t+1}$
 - Output relative to potential: $y_t - y_t^n$
 - Cost-push shocks: η_t
- Specific form above based on Calvo 83 sticky-price assumptions
- Details vary across specifications (e.g., sticky information yields $\bar{E}_{t-1} \pi_t$)
- Structural equation originating from firm's price setting decision

Estimating Slope of the Phillips Curve

$$\pi_t = \beta E_t \pi_{t+1} + \kappa(y_t - y_t^n) + \eta_t$$

Object of interest: Slope coefficient κ

- How much does an increase in “demand” / “tightness” / “output gap” affect inflation?

Tricky identification issues:

- Expected inflation unobserved
- “Natural rate of output” (i.e., supply shocks) unobserved
- Cost push shocks (e.g., variation in desired markups) unobserved

All three may cause omitted variables bias

Conventional Wisdom

- Volcker disinflation:
 - Tight policy -> high unemployment -> lower inflation
 - Substantial slope of the Phillips curve
- Since 1990:
 - Muted response of inflation to unemployment
 - Great Recession: missing disinflation
 - Late 2010s and 1990s: missing rise in inflation
- Phillips curve is getting flatter or hibernating
 - Perhaps an important flaw in the Keynesian model

Conventional Wisdom

- Assume adaptive expectations: $\beta E_t \pi_{t+1} = \pi_{t-1}$

- In this case,

$$\pi_t = \beta E_t \pi_{t+1} - \kappa(u_t - u_t^n) + v_t$$

becomes

$$\Delta \pi_t = -\kappa(u_t - u_t^n) + v_t,$$

- Stock and Watson (2019):
 - $\Delta \pi_t$: Annual change in 12-month core PCE inflation
 - $u_t - u_t^n$: CBO unemployment gap
 - Refer to κ as “Phillips correlation”

Flattening Phillips Curve

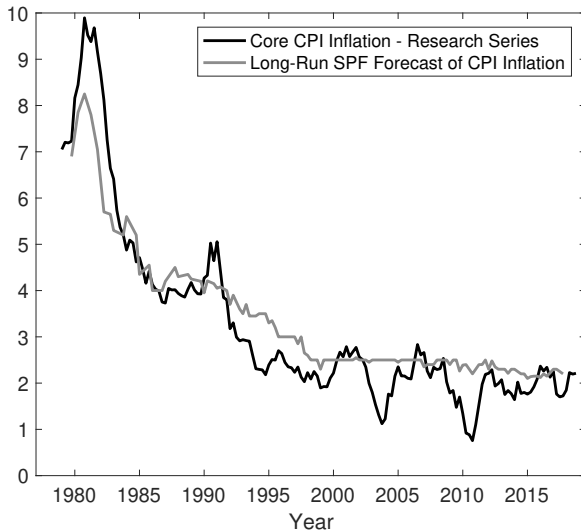


Figure 1: Stock and Watson's Changing Phillips Correlation

Alternative Explanation

- Volcker disinflation:
 - Sharp regime shift
 - Rapid fall in long-run inflation expectations
 - Rapid fall in inflation
- Since 1990:
 - Long-run inflation expectations have become anchored
 - Consequently, inflation has become more stable
- Apparent “flattening” of Phillips curve due to anchoring of inflationary expectations (Bernanke, 2007; Mishkin, 2007)

Long-Run Inflation Expectations



Why Might Inflation-Output Correlation Have Fallen?

1. Nominal rigidity increased over time
 - As menu cost model would predict
 2. Inflation expectations became better anchored
 - Output gap and change in inflation expectations correlated in 1970s and 1980s
 - But stable afterward
 - Biases estimates of Phillips curve slope
-
- **Today's central identification challenge:**
 - How to distinguish between these stories?

Forecasting and the Phillips Curve

- Important distinction between forecasting and structural relationships
 - Phillips Curves are bad at forecasting inflation (e.g. Atkeson & Ohanian 2001)
- According to theory: Phillips curve not necessarily useful for forecasting
 - Phillips curve is a supply curve
 - Useful for forecasting only if and when:
 - Most variation due to demand
 - Inflation expectations stable
 - Clearly not true in 1970s and 1980s
- Same as any other market
 - Supply curve for oil not necessarily useful to forecast price of oil

Key Questions

- Has the Phillips curve flattened?
- Is there missing disinflation / reflation?
- Does “anchoring” of inflation expectations explain stability of inflation?

Key Questions

- Has the Phillips curve flattened?
- Is there missing disinflation / reflation?
- Does “anchoring” of inflation expectations explain stability of inflation?
- **What about inflation after the pandemic?**

Three Strands of Phillips Curve Literature

1. Time series:
 - 1.1 **Aggregate Variation with Rational Expectations estimation**
 - 1.2 Aggregate Variation with Survey Expectations
2. Cross-Sectional Variation

Time Series Estimation with Rational Expectations (1/3)

- Typical approach (e.g. Gali & Gertler 1999) interested in estimating

$$\pi_t = \beta E_t \pi_{t+1} + \lambda mc_t + \eta_t$$

- Theory implies that mc_t is the appropriate “forcing variable”
- Under certain assumptions in the NK model, either:
 1. $mc_t = \kappa x_t$ where $x_t = y_t - y_t^n$ denotes the output gap; or
 2. $mc_t = s_t$ where s_t is labor's share of income
- When using (1) detrend output (e.g. HP filter or quadratic)
 - To remove variation in the natural rate
 - No strong theoretical justification for this step

Time Series Estimation with Rational Expectations (2/3)

- What do Gali & Gertler and others do about expectations of inflation?
- They assume rational expectations
- Under this assumption, Phillips curve can be written

$$\pi_t = \beta \pi_{t+1} + \lambda mc_t + \varepsilon_{t+1} + \eta_t$$

where $\varepsilon_{t+1} = \beta E_t \pi_{t+1} - \beta \pi_{t+1}$ (i.i.d.)

- They furthermore make a very strong assumption: supply shock term is IID
- These strong structural assumptions allow them to use lagged variables as instruments (any variable dated at time $t - 1$ or earlier)

Time Series Estimation with Rational Expectations (3/3)

- Maintained assumptions:

$$\pi_t - \beta \pi_{t+1} - \lambda mc_t = \varepsilon_{t+1} + \eta_t$$

where $\varepsilon_{t+1} + \eta_t$ is an i.i.d. expectations error + supply shock and (by assumption) correlated with $t - 1$ variables

- Implies:

$$E_t\{(\pi_t - \beta \pi_{t+1} - \lambda mc_t)z_{t-1}\} = 0$$

where z_{t-1} is in the time t information set of agents

Empirical Specification

- Gali-Gertler use GMM with these orthogonality conditions

$$E_t\{(\pi_t - \beta\pi_{t+1} - \lambda s_t)z_{t-1}\} = 0$$

Sample period: 1960Q1-1997Q4

- s_t is the labor share (also consider output gap specifications)
- Instruments: Four lags of inflation, labor income share, output gap, long-short interest rate spread, wage inflation, and commodity price inflation (24 instruments)

Empirical Concerns

- Is ε_{t+1} really just an i.i.d. expectations error?
 - If assumptions needed for $mc_t = s_t$ don't hold, it's not
 - If expectations are not rational, it is not
- Why should supply shocks be IID?
 - Slow moving supply shocks correlated with past stuff
 - E.g. oil shocks, TFP shocks, financial frictions
- 24 instruments raises concerns about many-weak instruments
 - Many/Weak instruments issue is an overfitting issue in small samples
 - Using 24 relatively weak instruments leads to substantial finite sample bias

Results

- Reduced form estimation with labor share:

$$\pi_t = \underset{(0.012)}{0.023} s_t + \underset{(0.045)}{0.942} E_t \pi_{t+1}$$

- Coefficients have “correct sign” and “sensible” magnitude
 - (Results with output gap are harder to interpret)
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- Structural estimates:
 - “Sensible” estimates for β , around 0.92 – 0.96
 - Estimates of θ on the high end—imply price rigidity of 5-6 quarters

Inflation Inertia

- Does NK Phillips curve account for inflation inertia?
- Gali-Gertler then estimate specification with fraction of rule-of-thumb agents
- Rule-of-thumb agents set

$$p_t^b = \bar{p}_{t-1}^* + \pi_{t-1}$$

This yields

$$\pi_t = \lambda mc_t + \gamma_f E_t \pi_{t+1} + \gamma_b \pi_{t-1}$$

where γ_f is decreasing and γ_b is increasing in ω (fraction of rule-of-thumb agents)

Estimation of Hybrid Phillips Curve

- Estimate of ω statistically significant, and in range $\omega = 0.265 - 0.486$
- A quarter to half of agents are rule-of-thumb
- Gali-Gertler conclusion:
 - Forward-looking behavior more important than backward-looking behavior
- Estimates of β on the low side at around 0.9

Critiques of Gali-Gertler

- Subsequent work: Gali-Gertler's results sensitive to specification issues
- Mavroeidis-Plagborg-Moller-Stock 14: main problem is weak instruments
 - Inflation is notoriously difficult to forecast
 - Lagged variables weak instruments for future inflation
 - **This is a problem for all rational expectations estimation of Phillips Curve**

Mavroeidis, Plagborg-Moller, Stock 14

- Run huge number of different a priori reasonable specifications with a common dataset
- Main findings:
 - Large amount of specification uncertainty
 - Large amount of sampling uncertainty
- Both conclusions due to weakness of identification

TABLE 4
NKPC SPECIFICATION COMBINATIONS

Specification settings	Options
Inflation (π_t)	GDP deflator, CPI, chained GDP def., GNP def., chained GNP def., NFB GDP def., PCE, core PCE, core CPI, filtered GDP def. gap, smoothed GDP def. gap, filt. CPI gap, sm. CPI gap, SPF-based CPI gap, filt. core CPI gap, sm. core CPI gap, filt. PCE gap, sm. PCE gap, filt. core PCE gap, sm. core PCE gap
Labor share (ls)	NFB, NFB coint. relation, HP filtered NFB gap, Baxter-King filt. NFB gap, linearly detrended NFB gap, quadratically detrended NFB gap, real-time NFB HP gap, real-time NFB BK gap, real-time NFB lin. detr. gap, real-time NFB quadr. detr. gap
Output gap (ygap)	CBO, HP filt., BK filt., lin. detr., quadr. detr., real-time HP filt., real-time BK filt., real-time lin. detr., real-time quadr. detr.
Reduced form	Unrestricted, VAR
Survey forecasts (π_{it}^s)	SPF CPI, SPF GDP def., GB GDP def.
Expectations	π_{t+1} (endogenous), $\pi_{t+1 t}^s$ (endog.), $\pi_{t+1 t}^s$ (exogenous) $\pi_{t+1 t-1}^s$ (endog.), $\pi_{t+1 t-1}^s$ (exog.)
Instruments	GC: 4 lags of π_t , ls, ygap, 10y-90d yield spread, wage infl., commodity price infl. GGLS: 4 lags of π_t and 2 lags of ls, ygap, wage infl. small: 4 lags of π_t and 3 lags of forcing variable exact: 1 extra lag of each endog. regr. (just-identified) RT: 2 real-time lags of GDP def. inflation, Δ ls, ygap survey: 2 lags of 1-quarter SPF/GB forecasts, forcing variable Extra regressors (e.g., oil) added to instruments (if endog., use 2 lags)
Inflation lags	0 lags (pure NKPC), 1 lag, 4 lags
Parameter restrictions	No restrictions, $\gamma(1) = \gamma$ (inflation coefficients sum to 1) With $\gamma(1) = \gamma_f$, use lags of $\Delta\pi_t$ instead of π_t as instruments
Oil shocks	None, log change of WTI spot price divided by GDP def.
Interest rate	None, 90-day Treasury rate
Sample	Full available, 1960-1997, 1968-2005, 1968-2008, 1971-2008, 1981-2008, 1984-end of sample
GMM estimator	2-step, CUE

Notes: List of the specification options that we consider when estimating the NKPC (9). The efficient GMM weight matrix is computed using the Newey and West (1987) heteroskedasticity and autocorrelation consistent estimator with automatic lag truncation, except for VAR specifications, which use the White (1980) heteroskedasticity consistent estimator.

Specification Uncertainty

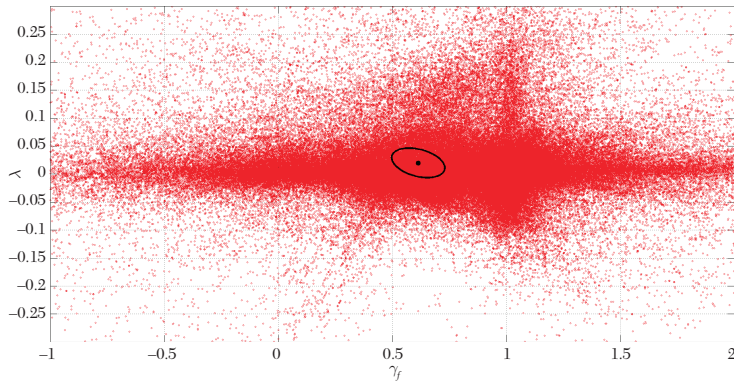


Figure 4. Point Estimates: Labor Share Specifications

Notes: Point estimates of λ , γ_f from the various specifications listed in table 4 that use the labor share as forcing variable, excluding real-time and survey instrument sets. The black dot and ellipse represent the point estimate and 90 percent joint Wald confidence set from the 1998 vintage results in table 3.

Mavroeidis, Plagborg-Moller, Stock 14

Overall conclusion:

“Literature has reached a limit on how much can be learned about the New Keynesian Phillips curve from aggregate macroeconomic time series.”

“New identification approaches and new datasets are needed to reach an empirical consensus.”

(Note: their critique applies more to rational expectations than survey expectations)

Three Strands of Phillips Curve Literature

1. Time series:
 - 1.1 Aggregate Variation with Rational Expectations estimation
 - 1.2 **Aggregate Variation with Survey Expectations**
2. Cross sectional / regional estimation

Ball and Mazumder (2019)

- Empirical specification:

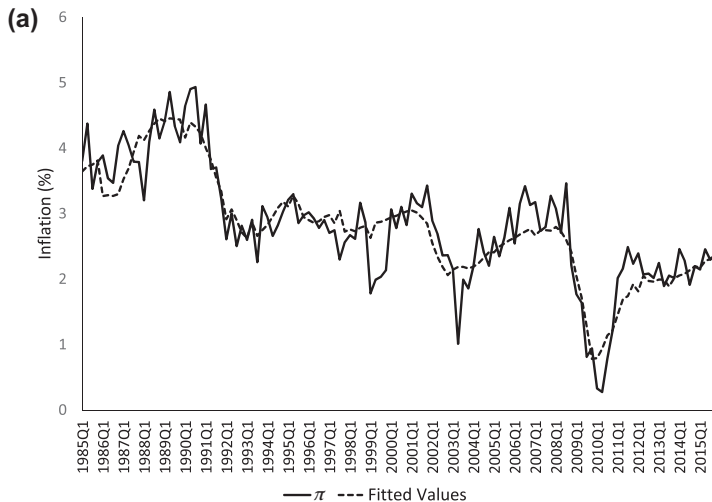
$$\pi_t = \pi_t^e + \alpha(u_t - u_t^*) + \varepsilon_t$$

- Focus on post-1985 period
- Main strength: use **long-run inflation expectations from survey data**
(see also Fuhrer & Moore 1995)
- Ignore endogeneity / supply shocks

Long-Run Inflation Expectations?

- Which inflation expectations should be used?
- Ball and Mazumder (2019) use long-run SPF inflation forecasts
 - Avoids rational expectations / weak instrument problem
- Doesn't New Keynesian model say one should use one-period-ahead inflation expectations?
- Can one just pick whatever one wants?
- We will come back to this (when discussing cross-sectional papers)

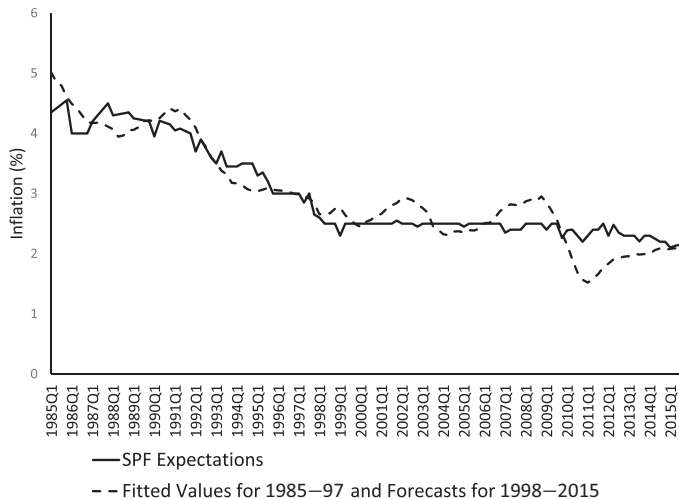
Good Fit! Stable, Including Great Recession



Anchoring of Inflation Expectations



Counterfactual with No Anchoring



- Focus on “missing disinflation” during Great Recession
- Argue that population explanations insufficient
 - Anchored inflationary expectations
 - Movements in natural rate
 - Flattening of the Phillips curve
- New explanation:
 - **Household inflation expectations** rose in 2009-2013
 - If firm's expectation the same, this can explain missing disinflation

$$\pi_t = \beta E_t \pi_{t+1} + \kappa(y_t - y_t^n) + \eta_t$$

Baseline assumptions:

- Assume $\beta = 1$
- Output gap measure: Unemployment rate

$$y_t - y_t^n = u_t$$

(Ignore natural rate u_t^n)

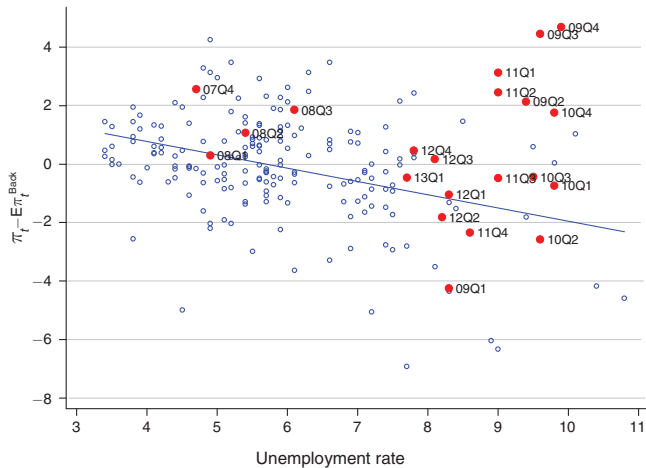
- Expectations of inflation, either:
 1. Backward looking $E_t \pi_{t+1}^{\text{Back}} = \pi_{t-1}$; or
 2. Measure of $E_t \pi_{t+1}^{\text{SPF}}$ from Survey of Professional Forecasters

$$\pi_t - E\pi_{t+1}^{\text{Back}} = \kappa u_t + \eta_t$$

- Estimate by OLS for sample 1960Q1-2007Q4
 - Implicitly assuming that $\eta_t \perp u_t$ (i.e., ignoring supply shocks)
- See whether Great Recession “sticks out”

Missing Disinflation

Panel A. CPI inflation and US unemployment



Similar results with $E_t \pi_{t+1}^{SPF}$

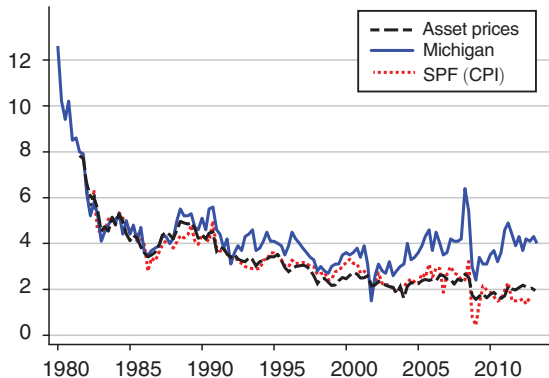
Source: Coibion and Gorodnichenko (2015)

New Explanation

- Expectations typically measured by SPF forecasts
- But is this the way to go?
- Right concept is **firms'** inflation expectations
 - Perhaps firm expectations exhibit similar biases to household expectations
- Show that household expectations (Michigan survey) have quite different properties from SPF
 - E.g. overreact to gasoline prices

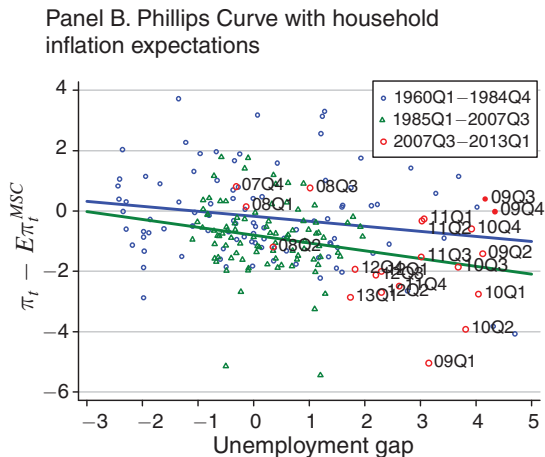
Household Expectations vs. SPF Forecasts

Panel A. Inflation expectations
for different economic agent



Source: Coibion and Gorodnichenko (2015)

No Missing Disinflation with Household Expectations



Also stable slope before vs. after 1985

Source: Coibion and Gorodnichenko (2015)

Key Questions Revisited

- Key questions:
 - Has the Phillips curve flattened?
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 - Does “anchoring” of inflation expectations explain stability of inflation?

Key Questions Revisited

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- Answers so far:

- Time series estimation w/ rational expectations not informative
- Time series estimation w/ survey data promising for stable Phillips Curve
- Caveats: limited variation, still sensitive to specification, omitted variable bias etc

The Need for Behavioral Macro

- Survey suggests significant departures from vanilla rational expectations:
 - (Un)anchoring important
 - Non-rational expectations potentially important during Great Recession
- **How do households form expectations?**
 - Limited micro data or macro modelling
 - Return to this at the end
 - “Gateway to behavioral macro”

McLeay & Tenreyro's Critique

- Silvana has a powerful critique of aggregate Phillips Curve estimation:
 - Suppose that monetary policy offsets demand shocks
 - Then variation in unemployment is due to supply shocks

- Simple example:

- Phillips Curve

$$\pi_t = \beta E_t \pi_{t+1} + \kappa \tilde{y}_t + \eta_t$$

- Monetary policy targets fixed inflation $\pi_t = \bar{\pi}$
 - Unemployment perfectly negatively correlated with supply shocks!

$$\bar{\pi} = \beta \bar{\pi} + \kappa \tilde{y}_t + \eta_t$$

$$\implies \tilde{y}_t = \frac{(1-\beta)\bar{\pi}}{\kappa} - \eta_t.$$

McLeay & Tenreyro's Critique

- Omitted variable bias problem will be severe
- Omitted variable bias reduces correlation between inflation and output
- Suppose policy has become better at offsetting demand shocks over time:
 - Then correlation between inflation and output falls

Three Strands of Phillips Curve Literature

1. Time series:

- Aggregate Variation with Rational Expectations estimation
- Aggregate Variation with Survey Expectations

2. Cross-Sectional Variation

- The aggregate data are not very informative
- So collect other sources of variation!

Can Cross-Sectional Data Help?

- Recent literature estimates “regional Phillips curves”
 - Fitzgerald-Nicolini 14; Kiley 15; Babb-Detmeister 17; McLeay-Tenreyro 19; Hooper-Mishkin-Sufi 19; Fitzgerald-Jones-Kulish-Nicolini 20; Beraja-Hurst-Ospina 19 (wages), Hazell-Herreno-Nakamura-Steinsson 21.
- Major advantages:
 - Fixed effects soak up variation in (common) **long-run inflation expectations**
 - Fixed effects soak up aggregate supply shocks
 - Better instruments to deal with regional supply shocks
- New challenge:
 - How is the slope of the regional Phillips curve related to the slope of the aggregate Phillips curve?

The Role of the Long-Run Inflation Target

- Let's understand better the central role of **long-run inflation expectations**:

$$\pi_t = \beta E_t \pi_{t+1} - \kappa(u_t - u_t^n) + v_t$$

- Solve forward:

$$\pi_t = -\kappa E_t \sum_{j=0}^{\infty} \beta^j u_{t+j} + \omega_t$$

where $\omega_t = E_t \sum_{j=0}^{\infty} \beta^j (\kappa u_{t+j}^n + v_{t+j})$.

- Looks like long-run inflation expectation vanishes due to discounting
- This is an illusion!

The Role of the Long-Run Inflation Target

- Useful to decompose u_{t+j} into permanent and transitory component:

$$\pi_t = -\kappa E_t \sum_{j=0}^{\infty} \beta^j u_{t+j} + \omega_t$$

becomes

$$\pi_t = -\kappa E_t \sum_{j=0}^{\infty} \beta^j \tilde{u}_{t+j} + \frac{\kappa}{1-\beta} E_t u_{t+\infty} + \omega_t$$

where $\tilde{u}_t \equiv u_t - E_t u_{t+\infty}$

- Since $\frac{\kappa}{1-\beta} E_t u_{t+\infty} = E_t \pi_{t+\infty}$, we have

$$\pi_t = -\kappa E_t \sum_{j=0}^{\infty} \beta^j \tilde{u}_{t+j} + E_t \pi_{t+\infty} + \omega_t$$

(Same result with $\beta = 1$)

The Role of the Long-Run Inflation Target

$$\pi_t = -\kappa E_t \sum_{j=0}^{\infty} \beta^j \tilde{u}_{t+j} + E_t \pi_{t+\infty} + \omega_t$$

- Long-run inflation target actually major determinant of current inflation
- Has a coefficient of one!!
- Current inflation moves one-for-one with beliefs about long-run inflation target
- In contrast, κ may be very small

The Role of the Long-Run Inflation Target

- To simplify, one can assume that $\tilde{u}_t$ follows an AR(1)
- This implies $E_t \tilde{u}_{t+j} = \rho_u^j \tilde{u}_t$

$$\pi_t = -\kappa E_t \sum_{j=0}^{\infty} \beta^j \tilde{u}_{t+j} + E_t \pi_{t+\infty} + \omega_t$$

becomes

$$\pi_t = -\psi \tilde{u}_t + E_t \pi_{t+\infty} + \omega_t$$

where $\psi = \kappa / (1 - \beta \rho_u)$.

The Role of the Long-Run Inflation Target

$$\pi_t = -\psi \tilde{u}_t + E_t \pi_{t+\infty} + \omega_t$$

- Variation in inflation may be dominated by variation in $E_t \pi_{t+\infty}$
- Variation in inflation may be **completely unrelated** to variation in $\tilde{u}_t$
- Worse still, correlation between $E_t \pi_{t+\infty}$ and $\tilde{u}_t$ potentially a source of severe omitted variables bias

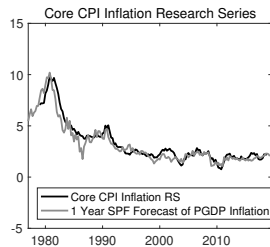
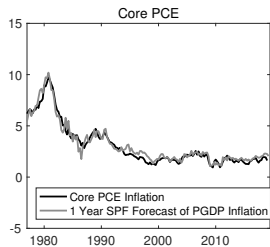
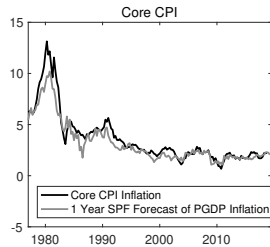
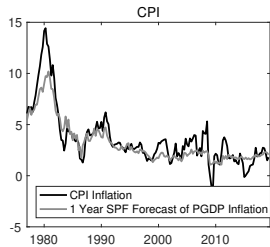
How Different Are π_t and $E_t\pi_{t+1}$?

$$\pi_t = \beta E_t\pi_{t+1} - \kappa(u_t - u_t^n) + v_t$$

- This is approximately (if $\beta \approx 1$):

$$\pi_t - E_t\pi_{t+1} \approx -\kappa(u_t - u_t^n) + v_t$$

- So, standard analysis aims to explain $\pi_t - E_t\pi_{t+1}$
- But how much is there to explain?
- Let's look at the difference between π_t and $E_t\pi_{t+1}$ in the data



$$\pi_t \approx E_t \pi_{t+1}$$

$$\pi_t - E_t \pi_{t+1} \approx -\kappa(u_t - u_t^n) + v_t$$

- Inflation gap for core inflation is small throughout
(for core using modern methods)
- Consistent with a flat Phillips curve

However:

- Relies heavily on exact timing of New Keynesian Phillips curve
(Also subject to other concerns regarding aggregate Phillips curve estimation)

A Model of the Regional Phillips Curve

- Two regions: Home and Foreign
- Tradeable and non-tradeable sector in each region
- No labor mobility between regions
- Perfect labor mobility between sectors within region
- Monetary union

Households and Firms

- Households:

- Consume and supply labor
- Nested CES demand over varieties of traded and non-traded goods
- GHH preferences

- Firms:

- Linear production function in labor
- Calvo (1983) type price rigidity

Phillips Curves

- Regional Phillips Curve for Non-Tradeables:

$$\pi_{Ht}^N = \beta E_t \pi_{H,t+1}^N - \kappa \hat{u}_{Ht} - \lambda \hat{p}_{Ht}^N + v_{Ht}^N$$

- Aggregate Phillips Curve:

$$\pi_t = \beta E_t \pi_{t+1} - \kappa \hat{u}_t + v_t$$

where $\hat{u}_{Ht} = -\hat{n}_{Ht}$ and $\hat{u}_t = -\hat{n}_t$

- Important result: Same slope κ
 - This is true for non-tradeable regional Phillips curve
 - Not for overall regional Phillips curve (traded goods priced nationally)
 - Relies on GHH preferences

Regional Phillips Curve for Non-Tradeables

$$\pi_{Ht}^N = \beta E_t \pi_{H,t+1}^N - \kappa \hat{u}_{Ht} - \lambda \hat{p}_{Ht}^N + v_{Ht}^N$$

- Extra term: $\lambda \hat{p}_{Ht}^N$. Theoretically important!
- Common critique:
 - Even in multi-region RBC model, regional demand shock would result in an increase in relative price of local goods
 - “Balassa-Samuelson Effect”
- Extra term implies that this model nests multi-region RBC model
- If prices were flexible, λ would be large
- Empirically, λ estimated to be small

Regional Phillips Curve Solved Forward

- Let's solve the regional Phillips curve forward:

$$\pi_{Ht}^N = -\kappa E_t \sum_{j=0}^{\infty} \beta^j \tilde{u}_{H,t+j} - \lambda E_t \sum_{j=0}^{\infty} \beta^j \hat{p}_{H,t+j}^N + E_t \pi_{t+\infty} + \omega_{Ht}^N,$$

- Long-run inflation expectations are constant across regions and can be replaced with time and state fixed effects:

$$\pi_{Ht}^N = -\kappa E_t \sum_{j=0}^{\infty} \beta^j \tilde{u}_{H,t+j} - \lambda E_t \sum_{j=0}^{\infty} \beta^j \hat{p}_{H,t+j}^N + \alpha_i + \gamma_t + \omega_{Ht}^N,$$

- Panel specification “differences out” long-run inflation expectations

Common Across Regions?

$$\pi_{Ht}^N = -\kappa E_t \sum_{j=0}^{\infty} \beta^j \tilde{u}_{H,t+j} - \lambda E_t \sum_{j=0}^{\infty} \beta^j \hat{p}_{H,t+j}^N + E_t \pi_{t+\infty} + \omega_{Ht}^N,$$

- Can't long-run inflation expectation differ across regions?
 - Prices are rising in New York relative to Kansas
 - Balassa-Samuelson effects
- Constant differences captured by state fixed effects
- Non-constant differences in **long-run** inflation expectations will be in error term
 - Small part of total variation (arguably)
 - A concern if correlated with instruments

Interpretation of Slope Coefficient

- Regional Phillips curve:

$$\pi_{Ht}^N = -\kappa E_t \sum_{j=0}^{\infty} \beta^j \tilde{u}_{H,t+j} - \lambda E_t \sum_{j=0}^{\infty} \beta^j \hat{p}_{H,t+j}^N + \alpha_i + \gamma_t + \omega_{Ht}^N,$$

- Suppose we assume that $\tilde{u}_{Ht}$ and $\hat{p}_{Ht}^N$ follow AR(1) processes:

$$\pi_{Ht}^N = -\psi \tilde{u}_{Ht} - \delta \hat{p}_{Ht}^N + \alpha_i + \gamma_t + \omega_{Ht}^N \quad (1)$$

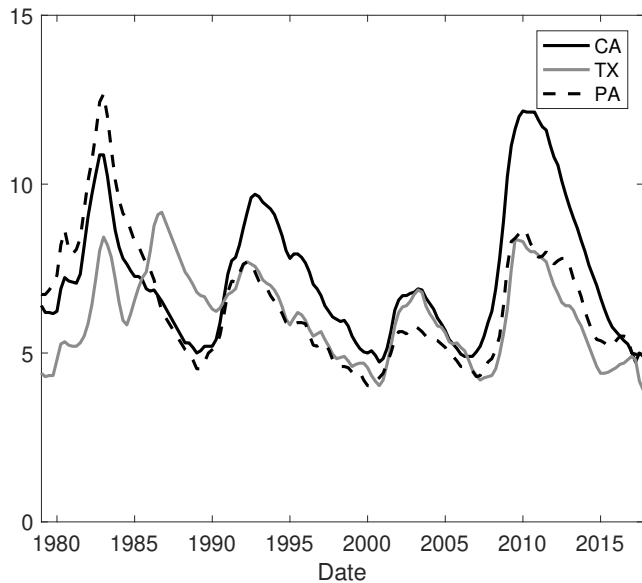
$$\text{where } \psi = \frac{\kappa}{1 - \beta \rho_u} \quad \text{and} \quad \delta = \frac{\lambda}{1 - \beta \rho_{pN}}$$

- Equation (1) similar to typical regional empirical specification
- But κ and ψ are not the same!
 - ψ potentially much larger than κ since $\tilde{u}_{Ht}$ is persistent
 - Prior regional Phillips curve literature estimates ψ not κ .
 - Helps explain large slope estimates in this literature

Data

- New state-level inflation indexes (Hazell-Herreno-Nakamura-Steinsson 21)
 - Sample period 1978 - 2018, quarterly
 - Based on BLS CPI micro data
 - Free of cross-state imputations
 - Separate indexes for tradeables vs. non-tradeables
- Analyze housing separately
- Measure of slack: State unemployment rates
- Tradeable demand spillover instrument:
 - State-industry employment shares
 - 2-digit SIC for 1975-2000
 - 3-digit NAICS from 1990-2018

Regional Business Cycles



Phillips Curve Slope

- Regional Phillips curve from model:

$$\pi_{it}^N = \alpha_i + \gamma_t - \kappa E_t \sum_{j=0}^{\infty} \beta^j u_{i,t+j} - \lambda E_t \sum_{j=0}^{\infty} \beta^j \hat{p}_{i,t+j}^N + \omega_{it}$$

- Reduced form equation similar to prior literature:

$$\pi_{it}^N = \alpha_i + \gamma_t - \psi u_{i,t-4} - \delta p_{i,t-4}^N + \varepsilon_{it}$$

- HHNS present estimates of both κ and ψ

Estimation of κ

$$\pi_{it}^N = \alpha_i + \gamma_t - \kappa E_t \sum_{j=0}^{\infty} \beta^j u_{i,t+j} - \lambda E_t \sum_{j=0}^{\infty} \beta^j \hat{p}_{i,t+j}^N + \omega_{it}$$

- Replace expectations with realized values and expectation error and truncate the infinite sums:

$$\pi_{it}^N = \alpha_i + \gamma_t - \kappa \sum_{j=0}^T \beta^j u_{i,t+j} - \lambda \sum_{j=0}^T \beta^j \hat{p}_{i,t+j}^N + \omega_{it} + \eta_{it}$$

where η_{it} is an expectations error (and truncation error)

- κ can now be estimated using an IV regression (i.e., GMM)
- Calibrate $\beta = 0.99$

Identification

Two Approaches:

1. Use lagged unemployment and relative prices as instruments
 - Unemployment may reflect supply shocks
 - Time fixed effects capture national supply shocks
 - Identifying assumption: No relative change in restaurant technology in Texas vs. Illinois when Texas experiences a recession relative to Illinois
2. Tradeable demand instrument

Tradeable Demand Spillover Instrument

$$\text{Tradable Demand}_{i,t} = \sum_{x \in T} \bar{S}_{x,i} \times \Delta \log S_{-i,x,t}$$

- $\bar{S}_{x,i}$: Average employment share of industry x in state i over time
- $\log S_{-i,x,t}$: National employment share of industry x at time t
- Identifying assumption:
 - Supply shocks not correlated with **both** shifts $\Delta \log S_{-i,x,t}$ **and** shares $\bar{S}_{x,i}$
- Intuition:
 - Oil boom increases labor demand and wages in Texas
 - “Demand shock” for Texan restaurants
 - Oil boom does not differentially affect production technology for restaurants in Texas

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- **We will spend more time on shift share instruments in lecture 5**

Estimation of ψ

$$\pi_{it}^N = \alpha_i + \gamma_t - \psi u_{i,t-4} - \delta p_{i,t-4}^N + \varepsilon_{it}$$

Same two approaches:

- OLS
- Instrument for $u_{i,t-4}$ with tradeable demand instrument

Full Sample Results

	No State Effects (1)	No Time Effects (2)	Lagged u IV (3)	Tradeable Demand IV (4)
ψ	-0.103 (0.036)	0.017 (0.027)	0.112 (0.057)	0.339 (0.126)
κ	-0.0037 (0.0013)	0.0003 (0.0019)	0.0062 (0.0028)	0.0062 (0.0025)
State Effects		✓	✓	✓
Time Effects			✓	✓

Has the Phillips Curve Flattened?

	Lagged u IV		Lagged u IV		Tradeable Demand IV	
	No Time Fixed Effects		Time Fixed Effects		Time Fixed Effects	
	(1)	(2)	(3)	(4)	(5)	(6)
	Pre 1990	Post 1990	Pre 1990	Post 1990	Pre 1990	Post 1990
ψ	0.449 (0.063)	0.009 (0.025)				
κ	0.0278 (0.0025)	0.0002 (0.0017)				

All specifications include state fixed effects

Has the Phillips Curve Flattened?

	Lagged u IV No Time Fixed Effects		Lagged u IV Time Fixed Effects		Tradeable Demand IV Time Fixed Effects	
	(1)	(2)	(3)	(4)	(5)	(6)
	Pre 1990	Post 1990	Pre 1990	Post 1990	Pre 1990	Post 1990
ψ	0.449 (0.063)	0.009 (0.025)	0.198 (0.113)	0.090 (0.057)		
κ	0.0278 (0.0025)	0.0002 (0.0017)	0.0107 (0.0080)	0.0050 (0.0038)		

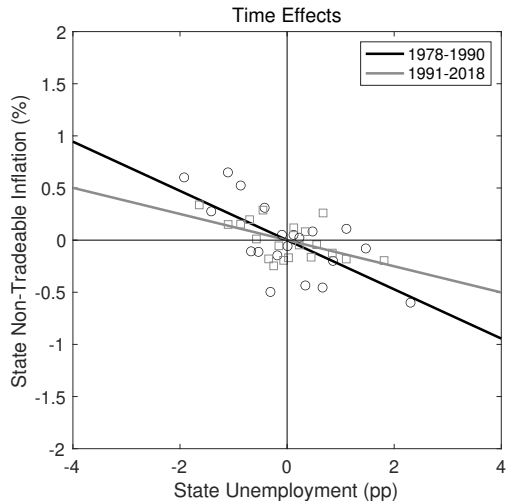
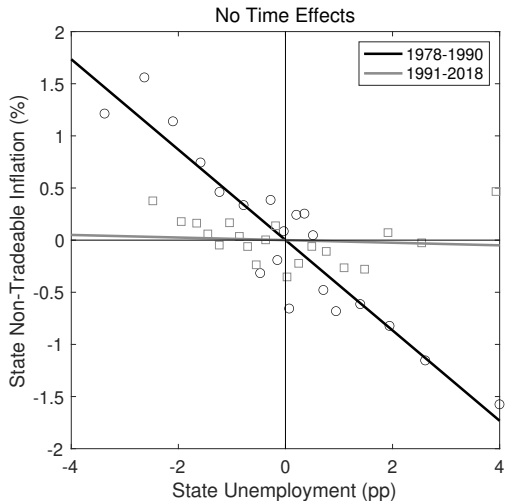
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	Lagged u IV No Time Fixed Effects		Lagged u IV Time Fixed Effects		Tradeable Demand IV Time Fixed Effects	
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	Pre 1990	Post 1990	Pre 1990	Post 1990	Pre 1990	Post 1990
ψ	0.449 (0.063)	0.009 (0.025)	0.198 (0.113)	0.090 (0.057)	0.422 (0.232)	0.332 (0.157)
κ	0.0278 (0.0025)	0.0002 (0.0017)	0.0107 (0.0080)	0.0050 (0.0038)	0.0109 (0.0048)	0.0055 (0.0029)

All specifications include state fixed effects

Scatterplots—Non-Tradeable Inflation and Unemployment



Main Conclusions

- Slope of Phillips curve small
 - $\kappa = 0.0062$ implies that even a 5 percentage point increase in unemployment decreases inflation by only 2 percentage points
 - If inflation expectations remain unchanged
- Apparent “flattening” mainly due to anchoring of expectations
 - No time fixed effects: Factor >100 flattening
 - With time fixed effects: Factor 2 flattening
 - Interpretation: Time fixed effects absorb movements in long-run inflation expectations

HHNS Estimates Compared to Prior Work

	κ
Gali (2008)	0.085
Rotemberg and Woodford (1997)	0.019
Nakamura and Steinsson (2014)	0.0077
Our Estimate	
Full Sample IV Estimate	0.0062

Note: HHNS adjust prior estimates by the elasticity of output with respect to employment in the model in these papers. For Nakamura and Steinsson (2014), HHNS use the calibration with GHH preferences.

Missing Disinflation?

- Can HHNS's cross-section estimate of κ explain aggregate time-series fluctuations in inflation?
- Many have argued:
 - Missing disinflation during Great Recession
 - Missing reflation during late 2010s and late 1990s
- Are cross-sectional estimates of Phillips curve steeper than time-series estimates?

Aggregate Implication

- Plot RHS and LHS of

$$\pi_t - E_t \pi_{t+\infty} = -\kappa \zeta \tilde{u}_t + \omega_t$$

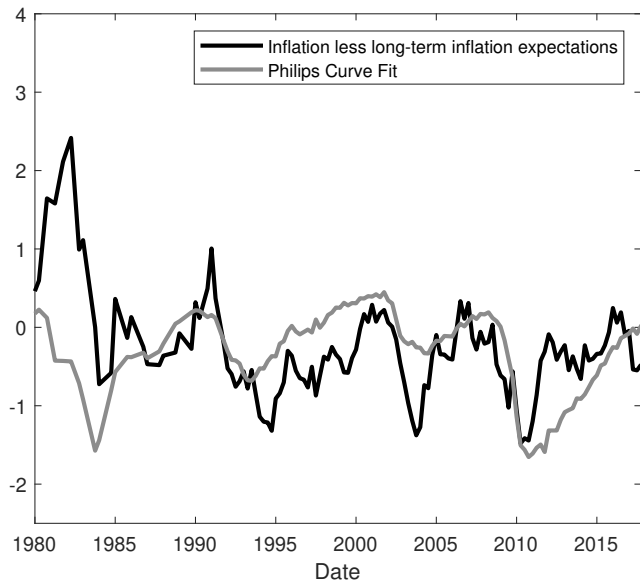
assuming no supply shocks $\omega_t = 0$

- Scaling factor: $\zeta = 6.16$ (s.e. 1.80)

$$\sum_{j=0}^T \beta^j \tilde{u}_{t+j} = \zeta \tilde{u}_t + \alpha + \varepsilon_t.$$

- Aggregate includes housing
 - Estimate aggregate Phillips curve for shelter
 - Data from American Community Survey for 2001-2017
 - $\kappa = 0.0243$ (s.e. 0.0053)
 - About four time larger than for non-shelter

Aggregate Implications



Has Phillips Curve “Broken Down” Recently?

- Post-1990: Predictions fit data reasonably well
 - Essentially no missing disinflation or missing reflation
- Pre-1990: Data deviates substantially from predictions
 - Actual inflation gap much higher than predicted
 - Natural Explanation: Adverse supply shocks
- Opposite of conventional wisdom

The Elephant in the Room

- Key determinant of inflation: $E_t \pi_{t+\infty}$
- But how does the monetary authority change $E_t \pi_{t+\infty}$
 - Fundamentally hard!!
 - How does it convince people that what it says is credible?
 - Answering this is not a strong suit of economists (need more research)
- Sometimes beliefs do change rapidly (e.g., Volcker disinflation, ends of hyperinflations)

How Does One Change Long-Run Beliefs?

- Volcker tightened policy dramatically
 - Caused massive recession
 - Didn't get fired
- Perhaps this was crucial in changing beliefs about long-run monetary regime
- Fundamentally different from view that inflation fell due to steep Phillips curve

Key Questions Revisited

- Key questions
 - Has the Phillips curve flattened?
 - Is there missing disinflation / reflation?
 - Does “anchoring” of inflation expectations explain stability of inflation?

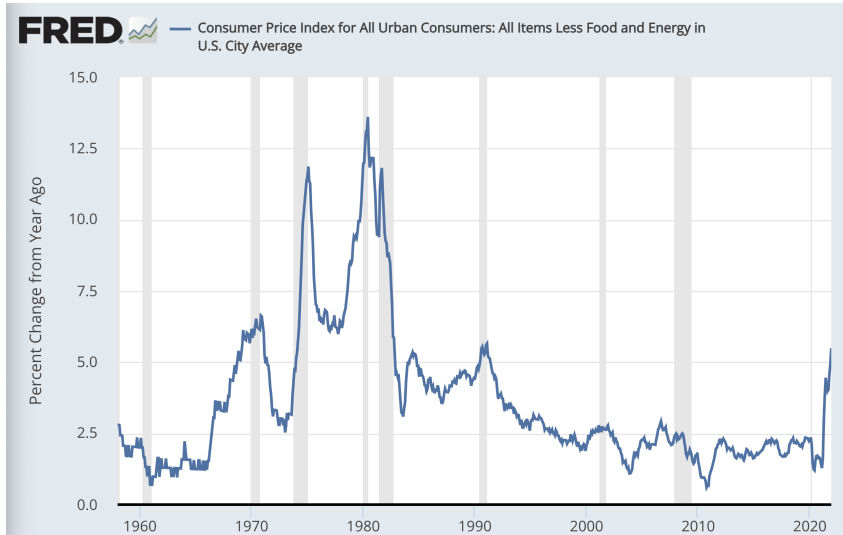
Key Questions Revisited

- Key questions *and answers from regional data*
 - Has the Phillips curve flattened? **No.**
 - Is there missing disinflation / reflation? **No.**
 - Does “anchoring” of inflation expectations explain stability of inflation? **Could be ...**

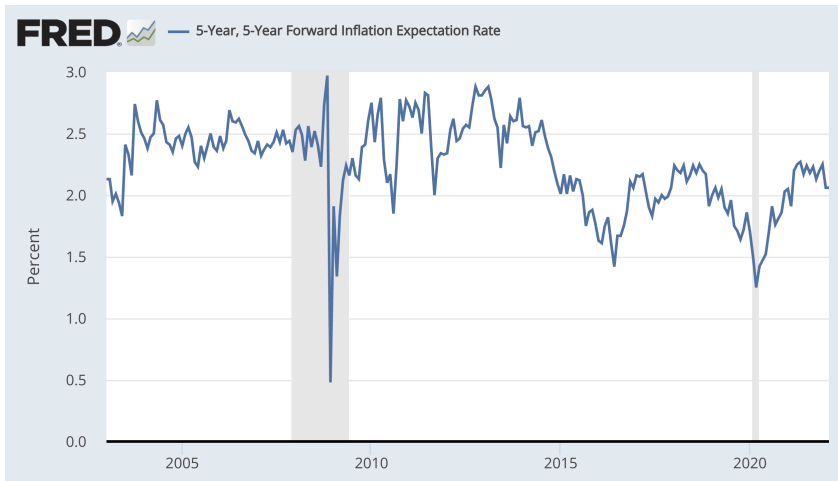
What About After the Pandemic?

- Stock and Watson (2010):
 - The history of the Phillips curve “is one of apparently stable relationships falling apart upon publication.”

Inflation After the Pandemic



Long Run Inflation Expectations Stable



Explanations

1. Nonlinear Phillips Curve? (Beaudry, Hou & Portier 2025)
2. Shocks + identification:
 - Fiscal policy (Hazell & Hobler 2025)

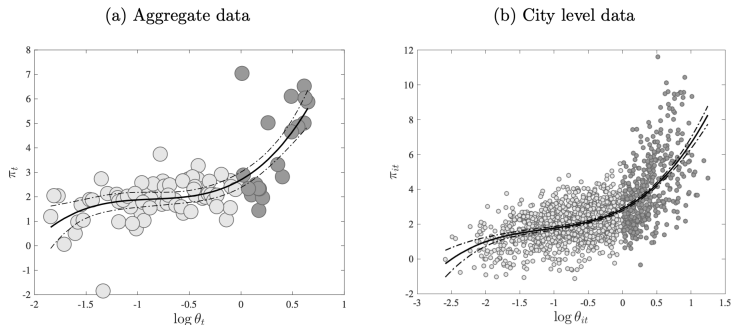
Nonlinear Phillips Curve?

An intuitive idea ...

$$\pi_t = \beta E_t \pi_{t+1} + \kappa(y_t - y_t^n) + \gamma \times I(\text{tight labor market}_t)(y_t - y_t^n) + \eta_t$$

Apparent Evidence of Nonlinearities ...

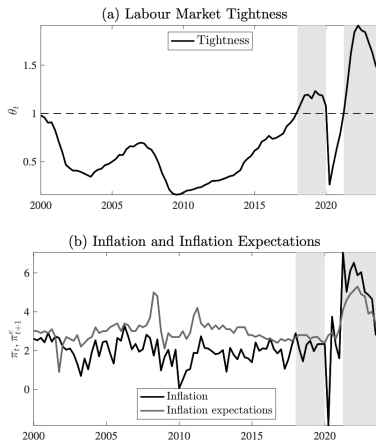
Figure 1: Labour Market Tightness and Inflation, Raw Data



Notes: Each dot represents a quarter (Panel (a)) or a quarter-city (Panel (b)). Labour market tightness is measured as $\log \theta$, where $\theta = V/U$. Inflation is quarter-to-quarter CPI core inflation (annualized). Light gray dots correspond to $\log \theta < 0$, dark gray dots correspond to $\log \theta > 0$. The black line is the fitted cubic relation between π and $\log \theta$, dotted lines delimit the 95% confidence interval. Sample is 2000Q1-2023Q4 for Panel (a) and 2000Q3-2024Q3 for Panel (b).

But Rising Inflation Expectations

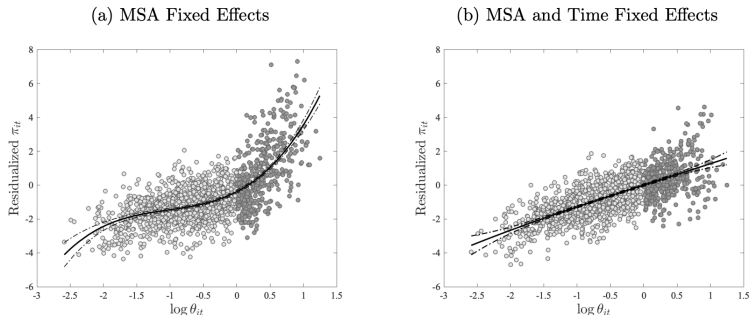
Figure 2: Labour Market Tightness, Inflation, Inflation Expectations



Notes: Labour market tightness is measured as $\log \theta$, where $\theta = V/U$. Inflation is quarter-to-quarter CPI core inflation (annualized). Expectations are one-year-ahead and obtained from the Michigan Survey of Consumers. Grey areas represent quarters with $\theta \geq 1$. Sample is 2000Q1-2023Q4.

No Non-Linearity with Time Fixed Effects

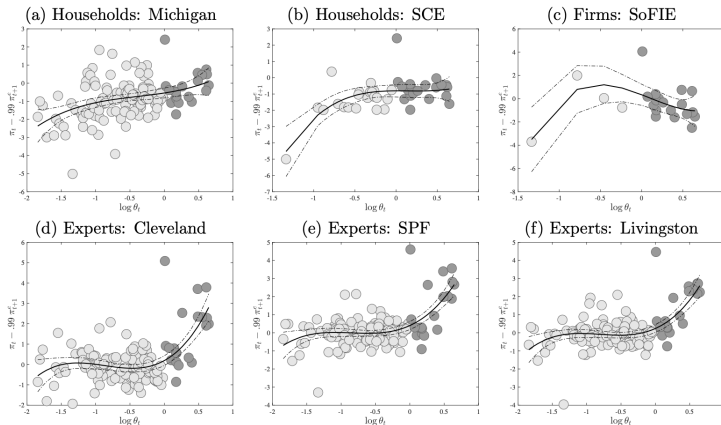
Figure 3: Inflation and Labour Market Tightness, City Level Data



Notes: Each dot represents a quarter-city. Dark dots indicate observations with $\log \theta_{it} \geq 0$ and light dots indicate observations with $\log \theta_{it} < 0$. In Panel (a), residualized inflation is obtained from the two-way linear fixed effects regression $\pi_{it} = \alpha_i + \kappa \log \theta_{it} + \varepsilon_{it}$ and computed as $\pi_{it} - \alpha_i$. In Panel (b), residualized inflation is obtained from the two-way linear fixed effects regression $\pi_{it} = \alpha_i + \gamma_t + \kappa \log \theta_{it} + \varepsilon_{it}$ and is computed as $\pi_{it} - \alpha_i - \gamma_t$. The black line is the fitted cubic relation between residualized π and $\log \theta$, dotted lines delimit the 95% confidence interval. Sample is 2000Q3-2024Q3.

Linearity in Aggregate (Expectations Measure Matters)

Figure 7: Inflation and Labour Market Tightness with Various Measures of Expectations



Explanations

1. Nonlinear Phillips Curve? (Beaudry, Hou & Portier 2025)
2. **Shocks + identification:**
 - Difficult to make progress with time series alone
 - One solution: look for different sources of identification and data
 - Fiscal policy (Hazell & Hobler 2025)

Do Deficits Cause Inflation?

Fiscal stimulus worth 13% of US GDP in Dec '20 and Mar '21, inflation rose to 8%

Empirical debate: some find deficits important for '20s inflation [e.g. Barro & Bianchi '23]

- Others—deficits not important [e.g. Bernanke & Blanchard '23]

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Empirical challenge: usual omitted variable concern [e.g. oil, supply bottlenecks, pandemic]

- Especially hard with single, episode specific shocks

Single episodes influential for macro theories [e.g. Great Depression, 60s-80s Inflation, Post Pandemic]

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This paper: high frequency narrative approach for causal effect of deficits on inflation

- Narrative measure of shock: deficit news event [Friedman & Schwarz '67; Romer & Romer '89]
- High frequency response: inflation expectations from asset prices [Gurkaynak et al '05]

Advantage: causal effect of single episode specific shocks

A High Frequency Narrative Approach: 2021 Georgia Senate Runoffs



A High Frequency Narrative Approach

Narrative identification of event: 2021 Georgia Senate Election Runoffs

- November 2020: Democrats win presidency + 48 senators
 - January 5th 2021: runoff election for 2 senators from Georgia
 - If Democrats win both: majority in senate + fiscal stimulus, small difference for non-fiscal
- Democrat victory = news about deficits

A High Frequency Narrative Approach

Narrative identification of event: 2021 Georgia Senate Election Runoffs

Narrative measure of shock: deficit news from Georgia

- New hand collected data: time-stamped information from 20 investment banks
 - $P(\text{Democrat victory}) \approx 50\%$, $\mathbb{E}[\text{Stimulus}] = \900 billion , 70% transfers, deficit financed
- Deficit news of \$450 bn = 2.1% of GDP
- Reports: main consequence of Democrat victory was fiscal stimulus

A High Frequency Narrative Approach

Narrative identification of event: 2021 Georgia Senate Election Runoffs

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High frequency response: inflation expectation from intraday swap prices

- Main identification strategy: single event study around election
- Advantage: excludes lower frequency omitted variables
- Result: expected price level increase of 0.38% over 2 years [persistent effect on inflation]
- Robustness: Capitol Hill Riots

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*Can **standard models** match the inflation multiplier?*

- **Simple HANK model** matches size + dynamics of post-Pandemic inflation multiplier
- Loose monetary policy important

Data

Intraday risk neutral inflation expectations

- 1, 2, 5 and 10 year inflation expectations from swaps (= contract on future inflation)
- Inflation risk premium is stable over this period [Cieslak & Pflueger '23, Cleveland Fed]

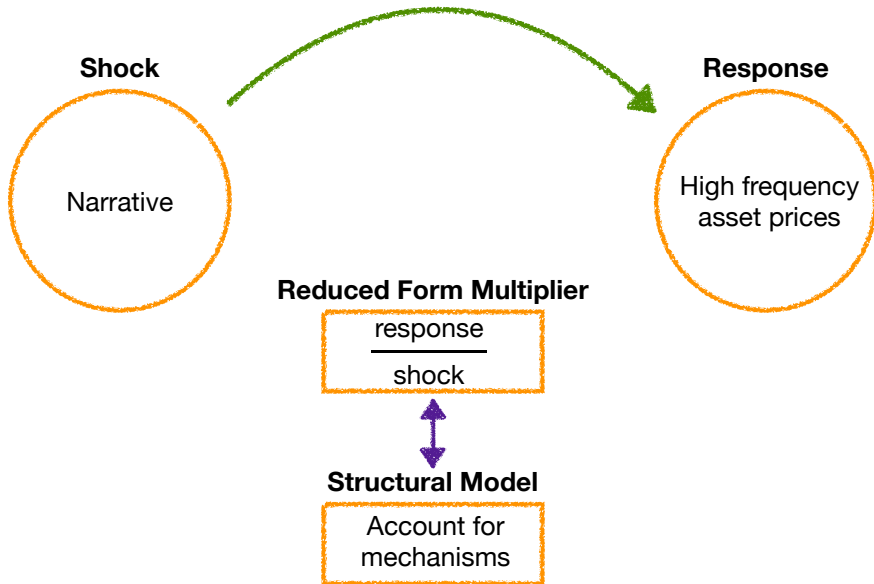
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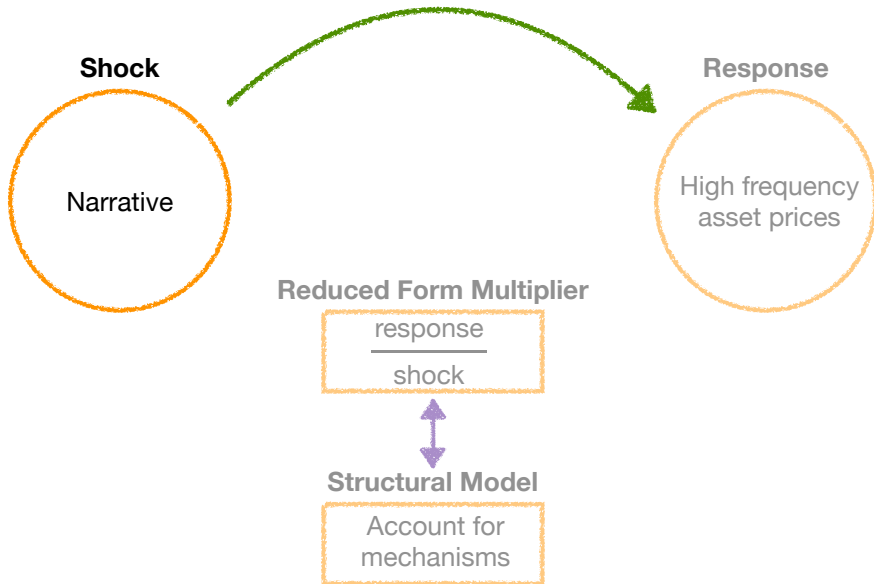
New hand collected narrative data from investment banks

- Source: investment banks + similar, 20 sources [e.g. Goldman Sachs, Bloomberg, Moody's]
 - Reports around events, distributed to market participants, time stamped, narrative detail
- Proxy for markets' beliefs

High Frequency Narrative Approach



High Frequency Narrative Approach



Narrative Identification of Event: 2021 Georgia Senate Election Runoffs

Mar '20: CARES Act \$2.2 trillion $\approx$ 10.3% of GDP

Nov '20: Biden elected, Democrats with 48 Senate seats

- 2 Georgia Senate seats to be decided in Jan 5

Nov '20 Election

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Democrat legislation:

- Mar '21: American Rescue Plan, \$1.8 trillion, $\approx 8.4\%$ of GDP
- Aug '22: approx. deficit-neutral Inflation Reduction Act

Nov '20 Election

Dec '20 Bipartisan Bill

Jan 5 '21 Georgia Runoff

Jan 6 '21 Capitol Hill Riots

Jan 7 '21 Democrats Declared Victors

March '21 American Rescue Plan

Aug '22 Inflation Reduction Act

Narrative Measure of Shock to News About Deficits

What was the shock to expected fiscal deficits due to Georgia runoff?

- **Challenge:** need **counterfactual** beliefs if **either** Democrats or Republicans were to win
- Search hand collected data from 20 investment banks, one week window around election

Needed: $\mathbb{E}[\text{stimulus} | \text{Democrat win}]$, $P(\text{Democrat win})$, $\mathbb{E}[\text{stimulus} | \text{Divided govt.}]$

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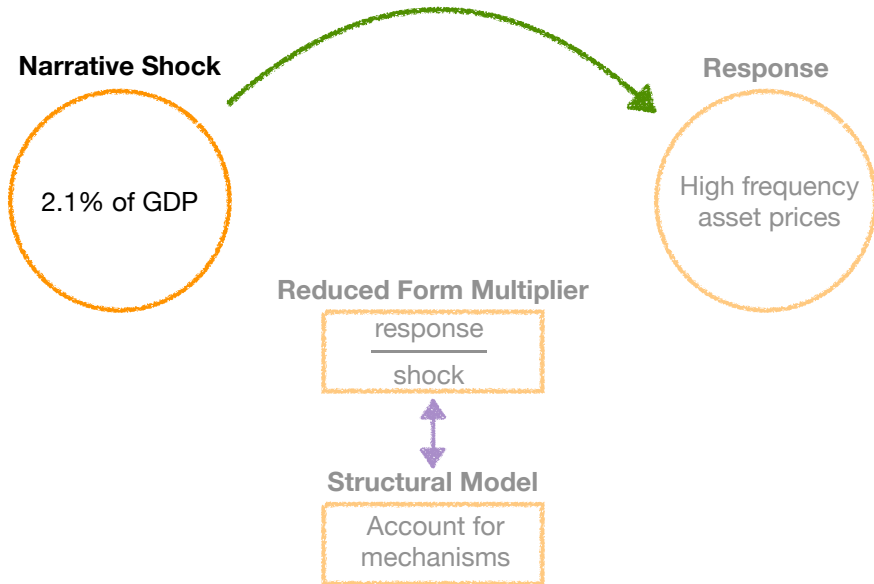
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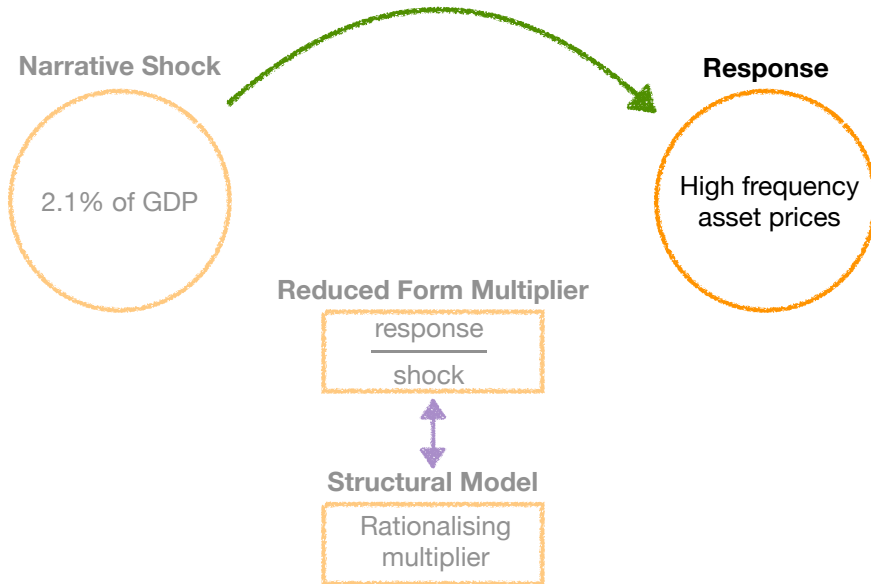
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Additional policy: delayed + tax financed infrastructure [quantitatively unimportant cf. Ramey '21]

High Frequency Narrative Approach



High Frequency Narrative Approach



High Frequency Response of Inflation Forecasts

- High frequency response of inflation expectations from swaps
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- **Identification assumption**: distribution of ε_t did not change just before vs. just after T
→ No other "atypical" shocks just after Senate election
... But typical shocks "allowed"

How Wide Should the Event Window Be?

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Potential confounding event: January 6th Capitol Hill Riots

- Robustness: end event window at 2PM on January 6th + other robustness to come
- Bloomberg daily “round up”: no other significant events until Jan 8 (payroll release)

▸ Details

Event Study: Causal Effect of Georgia Shock on Inflation Expectations



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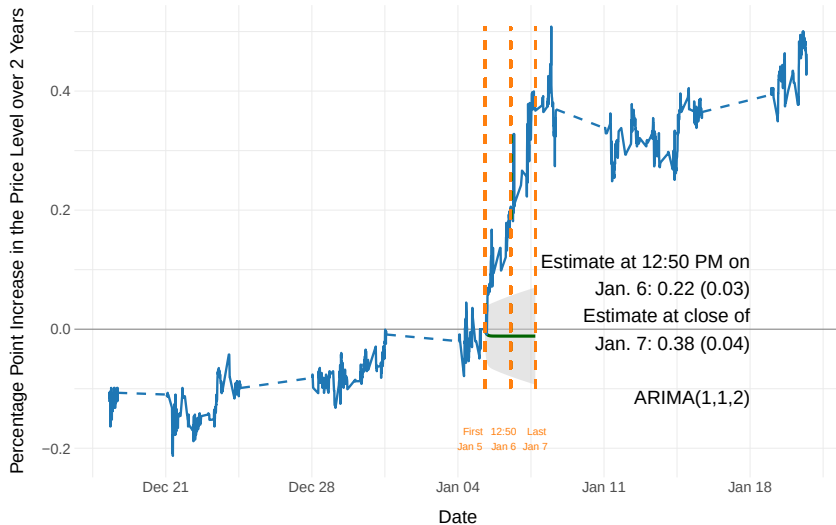
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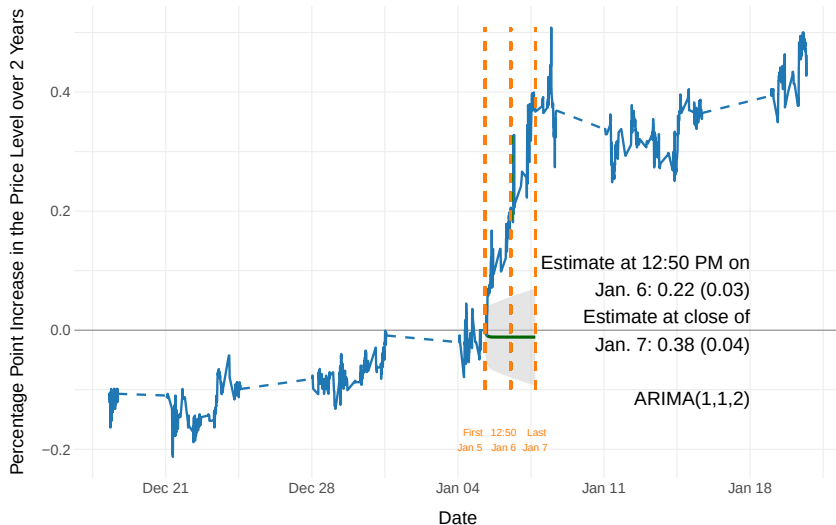
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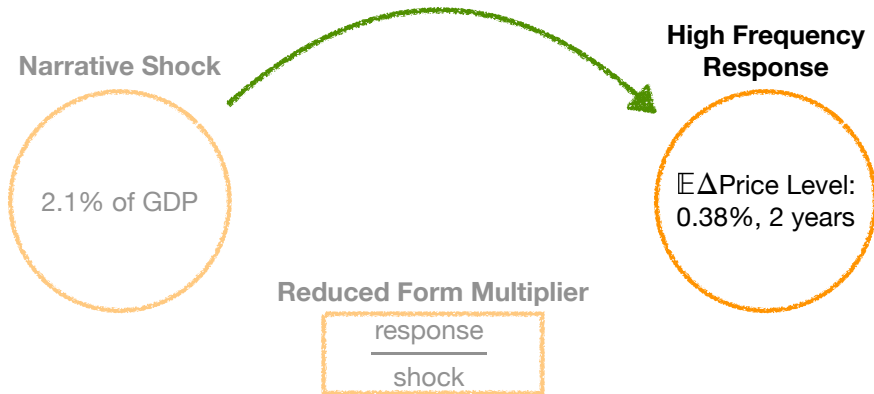
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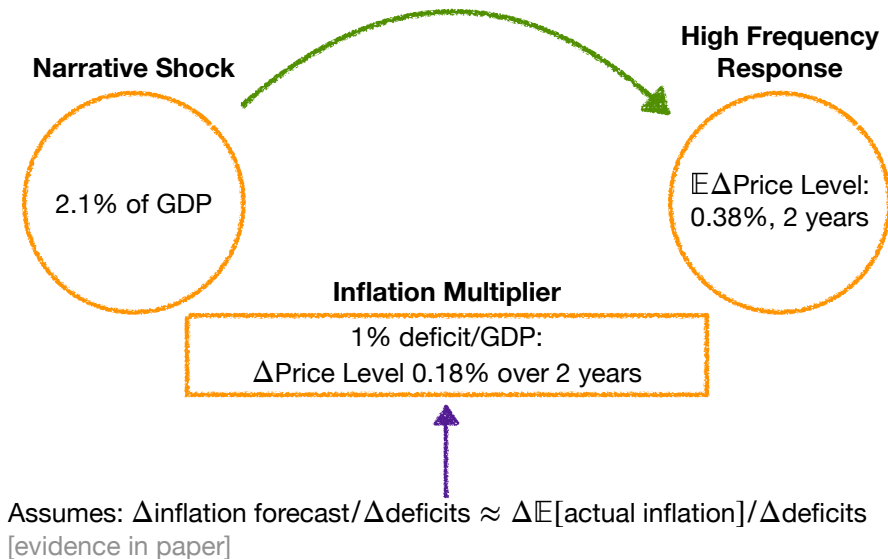


High Frequency Narrative Approach



Narrative + High Frequency: Effect of '21 Deficits on Inflation

Back of the Envelope: Combining Narrative + High Frequency



Inflation Multiplier

1% deficit/GDP:

Δ Price Level 0.18% over 2 years

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Extrapolation: Effect of Deficits on '21-'22 Inflation

13% of GDP [Dec '20 + Mar '21 stimulus] $\times$ 0.18 [inflation multiplier]
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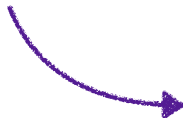
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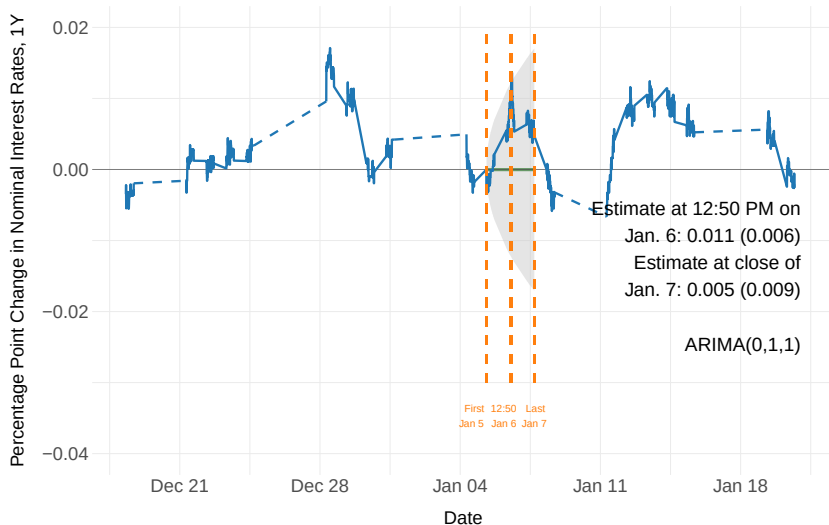
'21 Deficits + '20 CARES Act
explains 50% of '21-22 inflation

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Loose Monetary Policy Important: Short Term Rates Unchanged



But long term rates rise [cf. Mian, Straub & Sufi 2024]

► Long Term Rates + Robustness

► Fed Funds

Can Standard Models Match the Inflation Multiplier?

A Simple Heterogeneous Agent New Keynesian Model

Environment:

- Fraction μ **hand to mouth**: consume after tax income $C_{H,t} = W_t N_{H,t} - T_{H,t}$
- Fraction $1 - \mu$ of OLG households with survival probability ϕ

$$\max_{\{C_i, A_i\}} \mathbb{E}_t \sum_{j=0}^{+\infty} (\beta \phi)^j \left[\frac{C_{i,t+j}^{1-\frac{1}{\sigma}} - 1}{1 - \frac{1}{\sigma}} - \frac{N_{i,t+j}^{1+\frac{1}{\phi}}}{1 + \frac{1}{\phi}} \right] \quad \text{s.t.} \quad C_{i,t} + A_{i,t} = \frac{1}{\phi} \frac{1 + i_{t-1}}{\Pi_t} A_{i,t-1} + W_t N_{i,t} - T_{i,t} + Z_{i,t}$$

- Market clearing: $Y_t = C_t + G_t$
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Policy: government sets $\{i_t, T_t, T_t^H, G_t\}_{t=0}^{\infty}$ s.t. budget constraint

Calibration and Modelling '21 Deficit Shock

Standard calibration to **pre 2020 data**: [► Calibration Details](#)

- Calibrate consumption block to target intertemporal MPC [Fagereng et al '21; Auclert et al '23]
- **Flat but positively sloped Phillips Curve** [Hazell et al '22]

Main exercise: evaluating inflation response implied by model to “**deficit bundle**” from Georgia

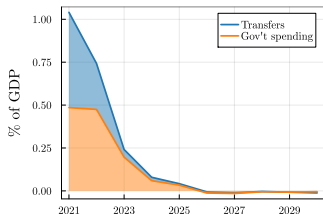
→ Compare to estimated inflation response

Measuring policy shock $\{i_t, T_t, T_t^H, G_t\}_{t=0}^{\infty}$ **around Georgia runoff**

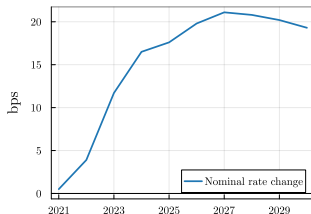
- **Deficit size + composition** (i.e. G vs. T) from narrative reports
- **Timing of spending** from realized American Rescue Plan
- **Debt repayment** from long term interest rates + Congressional Budget Office debt forecast
- **Nominal interest rate** shocks from high frequency response of interest rates
- Assume economy in steady state after 10 years

Model Quantitatively Matches Size + Persistence of Inflation Multiplier

Fiscal stimulus

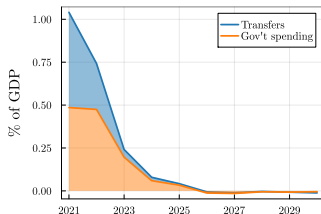


Monetary policy

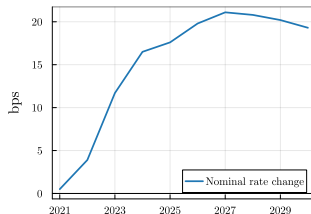


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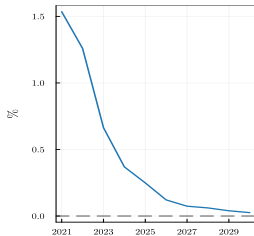
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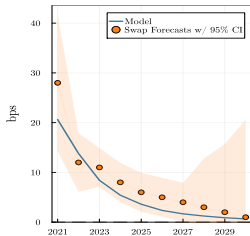
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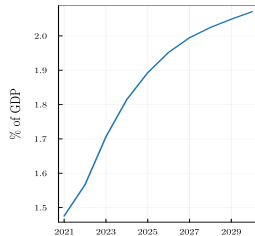
Output



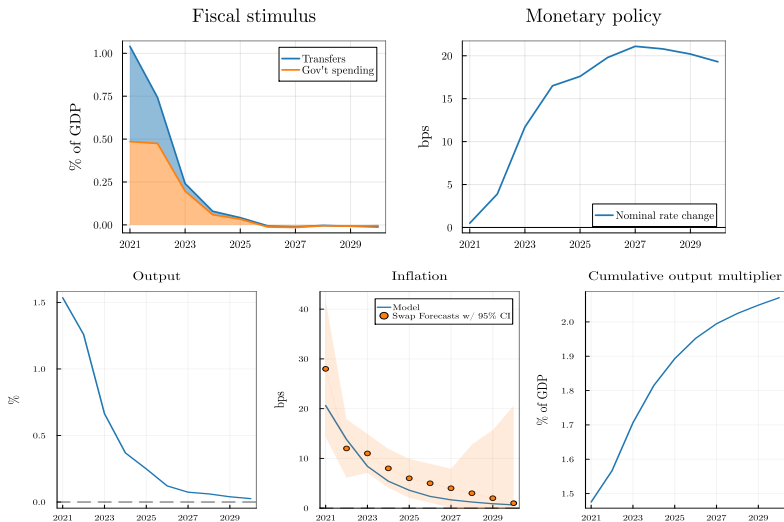
Inflation



Cumulative output multiplier



Model Quantitatively Matches Size + Persistence of Inflation Multiplier



In paper: key mechanism for inflation response is loose monetary policy

Conclusion

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This paper: **high frequency narrative approach** for causal effect of deficits on inflation

- **Narrative measure of shock**: deficit news from key event
- **High frequency response**: inflation expectations from asset prices

Bottom line:

- Dec '20 + Mar '21 stimulus explains **intermediate share** of '21-'22 inflation

→ Deficits **important** for inflation **but not only cause**

High frequency narrative approach estimates causal effect of **single episode specific** shocks

- In principle applicable to other influential episodes

Conclusion (1/2)

- Key questions:

- Has the Phillips curve flattened?
- Is there missing disinflation / re-inflation?
- Does “anchoring” of inflation expectations explain stability of inflation?

- Answers:

- Time series estimation w/ rational expectations not informative
- Survey data + cross-sectional estimates suggest:
 - Stable Phillips Curve w/ no missing disinflation
 - Gradual anchoring of inflation expectations

Conclusion (2/2)

- What about inflation after the pandemic?
 - Still great work to be done